

Black-Box Testing of Deep Neural Networks through Test Case Diversity

Aim of the Research Study:

The study investigates the effectiveness of black-box diversity metrics as an alternative to white-box coverage criteria for testing Deep Neural Networks (DNNs). The authors aim to address the limitations of existing coverage criteria, which require access to DNN internals or training data and often fail to correlate with fault detection in natural test input sets. The goal is to propose and validate black-box diversity metrics that can guide DNN testing more reliably and efficiently.

Methodology of the Study:

The authors selected and adapted three diversity metrics—Geometric Diversity (GD), Normalized Compression Distance (NCD), and Standard Deviation (STD)—to measure the diversity of test input sets. They used a pre-trained VGG-16 model to extract features from images and computed diversity scores. The study involved four datasets (Cifar-10, MNIST, Fashion-MNIST, SVHN) and five DNN models. A clustering-based approach was employed to estimate faults by grouping similar mispredicted inputs. The authors analyzed the correlation between diversity metrics, coverage criteria, and fault detection, and compared their computational performance.

Results:

The experiments revealed that Geometric Diversity (GD) outperformed other diversity metrics and state-of-the-art coverage criteria in terms of fault-revealing capability and computational efficiency. GD showed a consistent positive correlation with fault detection across all datasets and models, while coverage criteria exhibited weak or no correlation. The study also found that diverse input sets did not necessarily increase DNN coverage, challenging the assumptions underlying coverage-based testing.

Implications for Research and Practice:

For research, the study highlights the limitations of current coverage criteria and advocates for further exploration of black-box diversity metrics in DNN testing. For practice, the findings suggest that Geometric Diversity can serve as a reliable, model-independent metric to guide test case selection, minimization, or generation, especially in contexts where access to DNN internals is restricted. The results encourage a shift from white-box coverage to black-box diversity for more effective DNN testing.